

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:03-01-2025**
Time: 3HRS **GTM-10** **Max. Marks: 300**

PHYSICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

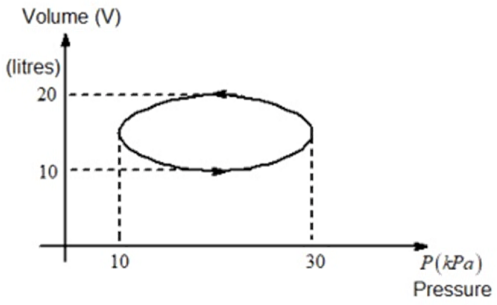
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The shortest wavelength associated with Balmer series is (given $R = 1.1\times10^{7}\,SI$ Unit)

   A) $3636A^{0}$ B) $912A^{0}$ C) $1818A^{0}$ D) $6568A^{0}$

2. The work done by a gaseous system in going through a cyclic elliptical process shown in the figure is



   A) $25\pi J$ B) $100\pi J$ C) $50\pi J$ D) $-50\pi J$

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3. The electric field of an electromagnetic wave is given by $E = 100\sin\omega\left(t - \frac{x}{v}\right) N/C.$

Find the energy contained in a hypothetical space of volume 100cm^3 (Given $\in_0 = 9\times10^{-12} C^2 N^{-1} m^{-2}$)

A) $4.5\times10^{-12} J$ B) $9\times10^{-12} J$ C) $2.25\times10^{-12} J$ D) $6\times10^{-12} J$

4. Find the difference of weight of a body from surface of earth to at a depth R/4 from the surface of earth. If it's weight on the surface of earth is 400 N

A) 100N B) 200N C) 300N D) 400N

5. The forbidden energy gap of a semiconductor material is 2.1 eV. The maximum wavelength of photon of sodium light is required to create a hole- electron pair is nearly (Take: $hc = 12400 eV\,\overset{0}{\text{A}}$)

A) $5904\,\overset{0}{\text{A}}$ B) $5450\,\overset{0}{\text{A}}$ C) $6000\,\overset{0}{\text{A}}$ D) $5500\,\overset{0}{\text{A}}$

6. A water film is made between two straight parallel wires of length 10 cm separated by 5 mm from each other. If the distance between the wires is increased by 2 mm. How much work will be done? Surface tension for water is 72 dyne cm^{-1}

A) 288 erg B) 72 erg C) 144 erg D) 216 erg

7. A bike of mass 300 kg taking turn on a banked road of radius 100m and banking angle 37^0 If co efficient of static friction is 0.25. Then the maximum speed with which car can negotiate the turn safely. ($g = 10 m/s^2$)

A) $28.5 m/s$ B) $35 m/s$ C) $40 m/s$ D) $42.5 m/s$

8. Two small identical conducting sphere A and B with equal and opposite charge Q on each other attract with a force 64N. A third identical uncharged conducting sphere is successively brought in contact with the two sphere and removed to infinity. The new force between the sphere A and B is

A) 16 N B) 32 N C) 8 N D) 12 N

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9. When 10 A current is flowing in an induction coil then the magnetic energy stored in the coil is 1J. The self-induction of the coil is

A) 5 mH B) 10 mH C) 15 mH D) 20 mH

10. A convex and concave lens of focal length 10 cm and 20 cm respectively are kept in contact with each other. Find focal length of the combination of lens

A) $\frac{20}{3}$ cm converging B) 20 cm converging

C) 20 cm diverging D) $\frac{20}{3}$ cm diverging

11. If the ratio of number of moles of hydrogen gas to oxygen gas is two, the ratio of total translational kinetic energy of hydrogen gas molecules to that of oxygen molecules at $20^0 C$ is

A) 1:1 B) 2:1 C) 1:8 D) 1:16

12. A body of weight 250N is hanging on a spring balance which is suspended from the ceiling of an elevator at rest. Now elevator starts accelerating upward with an acceleration $2m/s^2$. Then the reading of spring balance when body comes to an equilibrium relative to the elevator is $\left(g = 10m/s^2\right)$

A) 200 N B) 250 N C) 300 N D) 350 N

13. When a photon of wavelength 400nm is incident on a metal surface, the electron emission takes place. The stopping potential to stop electron is 1.5V. The work function of metal surface is: (given hc = 1240 eV-nm)

A) 1.5 eV B) 1.6 eV C) 2 eV D) 2.2 eV

14. The number of electrons flowing per second in the filament of a 100W bulb operating at 240V is (given charge of electron $e = 1.6 \times 10^{-19} c$)

A) 6.25×10^{18} B) 3.1×10^{17} C) 1.3×10^{18} D) 2.6×10^{18}

15. When the momentum of a body increased by 200% than its kinetic energy increased by

A) 400% B) 600 % C) 800 % D) 900 %

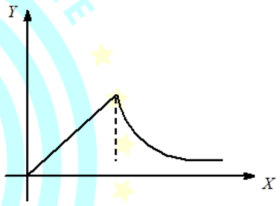
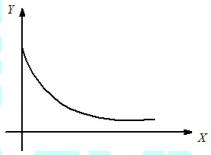
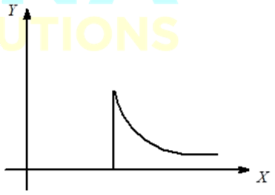
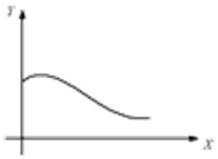
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16. Pressure inside a liquid drop is greater than the pressure outside by an amount $\frac{2S}{R}$, where S is surface tension of liquid and R is radius of liquid drop. When 8 such identical drops combine together become one single drop than the pressure difference between inside and outside the drop will be

A) $\frac{S}{R}$ B) $\frac{2S}{R}$ C) $\frac{4S}{R}$ D) $\frac{S}{4R}$

17. **Match List - I with List - II**

	List – I (Y vs X)		**List – II (Shape of graph)**
a)	Y= magnetic susceptibility X= absolute temperature	I)	
b)	Y= magnetic field X= distance from the Axis of current carrying Cylindrical pipe	II)	
c)	Y- magnetic field X- Distance from The axis of current Carrying cylindrical Conductor	III)	
d)	Y- magnetic field X- Distance from Centre of current carrying Circular ring along its axis	IV)	

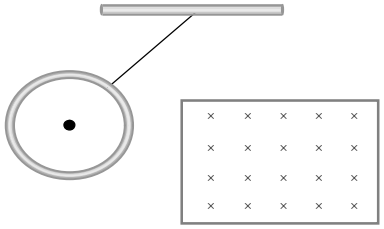
A) $(a)-II,(b)-III,(c)-IV,(d)-I$ B) $(a)-I,(b)-III,(c)-II,(d)-IV$

C) $(a)-II,(b)-III,(c)-I,(d)-II$ D) $(a)-III,(b)-II,(c)-IV,(d)-I$

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18. A metallic disc connected to a rod oscillates freely like a pendulum. If now a magnetic field is applied in horizontal direction so that the pendulum now swings through the field, the pendulum will



A) Keep oscillating with the old time period

B) Keep oscillating with a smaller time period

C) Keep oscillating with a larger time period

D) Continuously slows down and come to rest

19. Given below are two statements

Statement (I) : Dimensional formula of planck's constant is $[ML^2T^{-1}]$

Statement (II) : Dimensional formula of bulk modulus is $[ML^{-1}T^{-2}]$

A) Both the statement (I) and (II) are correct

B) Both the statement (I) and (II) are incorrect

C) Statement (I) is incorrect and statement (II) is correct

D) Statement (I) is correct and statement (II) is incorrect

20. A body projected vertically upwards with a certain speed from top of a tower, reaches the ground in time t_1. If it is projected vertically downwards with same speed from same point, it reach the ground in the time t_2. Time required for it to reach the ground if it dropped from the same point and projected horizontally with same speed, respectively are

A) $\sqrt{t_1+t_2}, \sqrt{t_1-t_2}$

B) $\sqrt{t_1+t_2}, \sqrt{t_1+t_2}$

C) $\sqrt{t_1 t_2}, \sqrt{t_1+t_2}$

D) $\sqrt{t_1 t_2}, \sqrt{t_1 t_2}$

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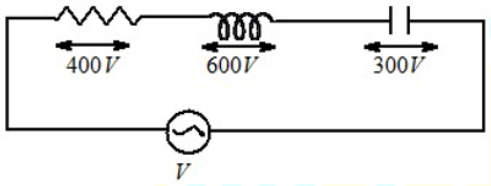
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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. In an Millikan's experiment an oil drop of radius $2\times10^{-6}m$, density $900kg/m^3$ is held stationary between two condenser plates 1.0 cm apart and kept at a potential difference of 1.5π kV. If air effect is neglected, then the number of excess electrons carried by oil drop _____ (Take: $g=10m/s^2$)

22. In a series LCR circuit as shown in the diagram. The supply voltage is ____Volt



23. A particle is moving in straight line, it's velocity (v) with time (t) is given as $v=\left(t^2-1\right)m/s$. The distance travelled by the particle between $t=0$ *to* $t=2$ sec. is _____ cm

24. Six particle each of mass 1 kg , 2kg , 3kg, 4kg, 5kg and 6kg are kept at the vertex of regular hexagon of side 1 m. The moment of inertia of system passing through its center and perpendicular to its plane is ____kgm^2

25. A circular coil having 50 turns and radius $\sqrt{\frac{7}{22}}m$ carrying current 1mA, is placed in uniform magnetic field 2.0T such that plane of coil is perpendicular to magnetic field . The work done in turning the coil through 180^0 is ____ $mJ\left(\pi=\frac{22}{7}\right)$

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CHEMISTRY **MAX.MARKS: 100**

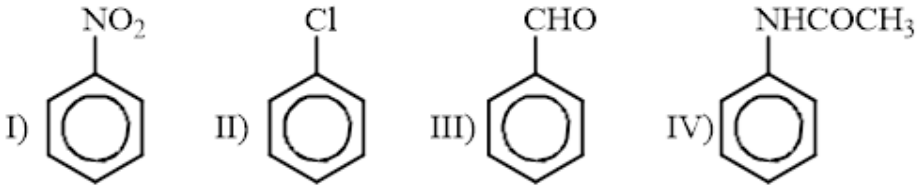
SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26.



The correct order of decreasing reactivity towards electrophilic substitution amongst the above given compounds is:

A) $(III)>(I)>(II)>(IV)$ B) $(IV)>(III)>(II)>(I)$

C) $(IV)>(II)>(I)>(II)$ D) $(IV)>(II)>(III)>(I)$

27. The molality of 1 litre solution of 93% $\left(\frac{w}{v}\right)$ H_2SO_4 solution is:

(Given: Density of solution = 1.84gm/ml)

A) 8.42m B) 11.57m C) 10.42m D) 9.98m

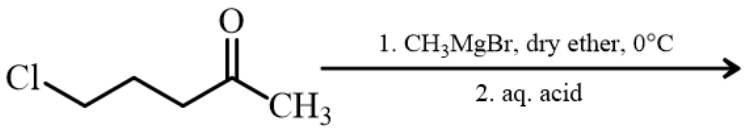
28. Match the Column-I with appropriable bonds given in Column-II

Column I		Column II	
a)	Polysaccharide	I)	H-bond
b)	Protein	II)	Ester bond
c)	Water	III)	Peptide bond
d)	Fat	IV)	Glycosidic bond

A) $a-IV, b-II, c-I, d-III$ B) $a-IV, b-III, c-I, d-II$

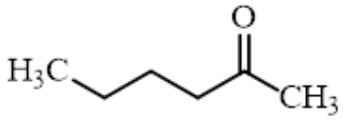
C) $a-III, b-IV, c-I, d-II$ D) $a-III, b-II, c-I, d-IV$

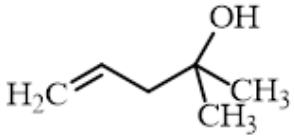
29. The major product in the following reaction is

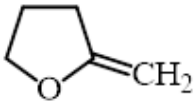


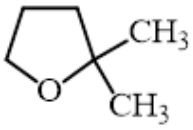
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A) 

B) 

C) 

D) 

30. A reddish brown gas, obtained on heating an inorganic compound with $K_2Cr_2O_7$ and conc. H_2SO_4 was bubbled through dil.NaOH. The alkaline solution yielded a Yellow precipitate on addition of lead acetate. The inorganic compound is likely to be a

A) Chloride B) Nitrate C) Bromide D) Sulphide

31. Which of the following statement is correct?

A) In a galvanic cell electrical energy produces a chemical reaction

B) Electro chemical cell can be converted into an electrolytic cell by reversing the flow of ions in salt-bridge

C) salt-bridge minimizes the liquid-liquid junction potential

D) An electro chemical cell can be converted into an electrolytic cell by applying an external opposite potential lower than E^0_{cell}

32. What is the correct order of spin only magnetic moment (in BM) of Mn^{+2}, V^{+2} and Cr^{+2}

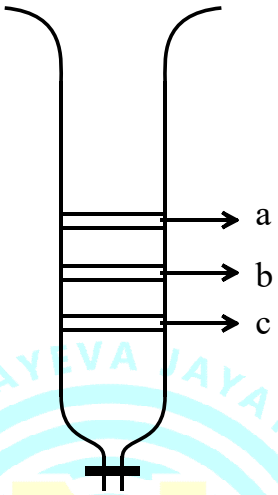
A) $Mn^{+2} > V^{+2} > Cr^{+2}$ B) $V^{+2} > Cr^{+2} > Mn^{+2}$

C) $Mn^{+2} > Cr^{+2} > V^{+2}$ D) $Cr^{+2} > V^{+2} > Mn^{+2}$

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33. From the figure of column chromatography given below, identify the CORRECT statement?



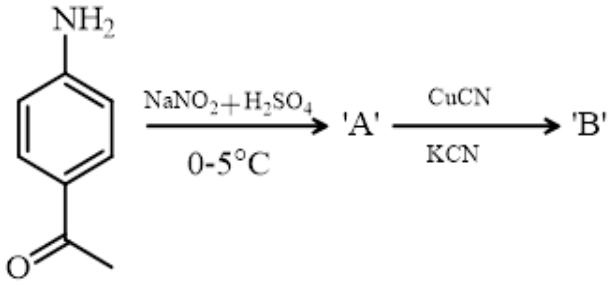
A) Compound ‘c’ is more polar than ‘a’ and ‘b’

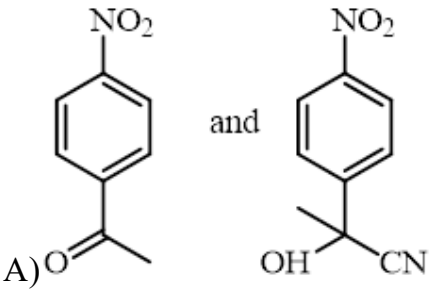
B) Compound ‘a’ is least polar

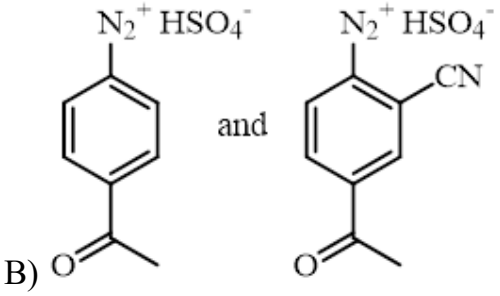
C) Compound ‘b’ is comes out of the column before ‘c’ and after ‘a’

D) Compound ‘a’ spends more time in the column

34. The major products ‘A’ and ‘B’ formed in the following reaction are respectively:

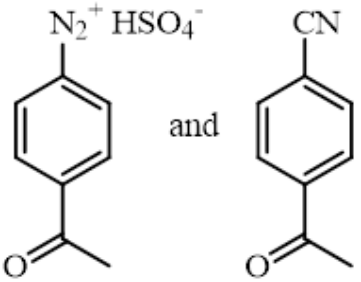


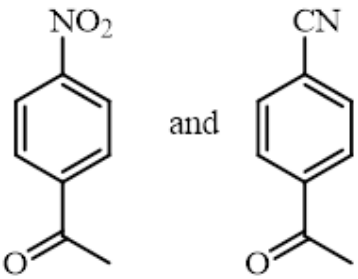
A) 

B) 

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C) 

D) 

35. The incorrect statement regarding the optical isomerism.

A) Racemic modification is an equimolar mixture of dextrorotatory and laevorotatory isomers

B) Meso compounds contain more than one chiral center

C) Meso compounds are externally compensated

D) Racemic mixture is designated as d*l*-pair

36. **Statement I**:- PF_3 and BrF_3 both have same geometry

Statement II:- Both XeF_4 and $[CoF_6]^{3-}$ central atom is sp^3d^2 hybridized

In the light of the above statements, choose the correct answer from the options given below

A) Statement I is true but Statement II is false

B) Both Statement I & Statement II are true

C) Both Statement I & Statement II are false

D) Statement I is false but Statement II is true

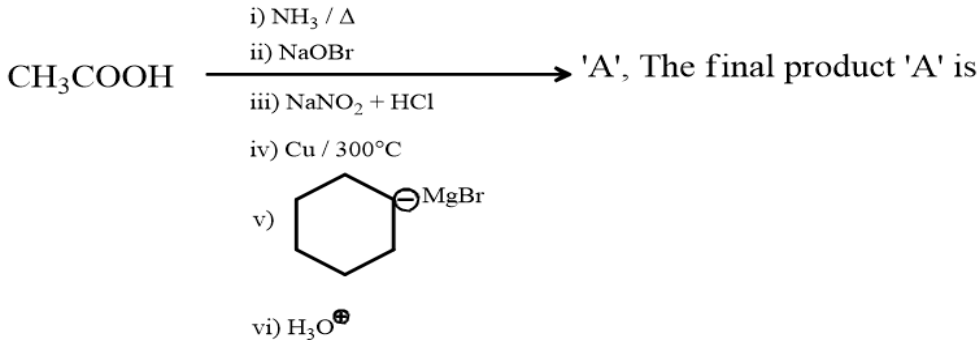
37. V_{ml} of 0.1M benzoic acid $(p_{Ka} = 4.2)$ solution is added into 0.2 M sodium benzoate solution and formed a 300ml resultant acidic buffer solution. If pH of this buffer solution is 4.5 then V_{ml} is ___ (Given: $\log 2 = 0.3$)

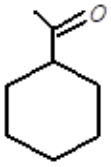
A) 100 ml B) 150 ml C) 200 ml D) 175 ml

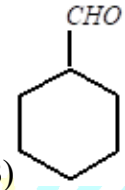
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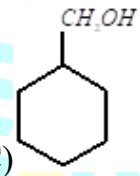
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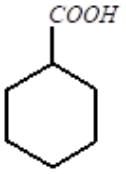
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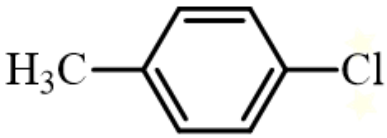
A) 

B) 

C) 

D) 

39. The IUPAC name(s) of the following compound is (are)



i)1-chloro-4-methylbenzene
ii) 4-chlorotoluene
iii) 4-methylchlorobenzene
iv 1-methyl-4-chlorobenzene

A) Only (i) is correct
B) Both (i) & (ii) are correct
C) Only (ii) is correct
D) Only (iii) is correct

40. For a first order reaction, among the following which is/are correct/s?

i) The degree of dissociation is equal to $\left(1-e^{-kt}\right)$

ii) The time taken for the completion of 87.5% reaction is twice the $t_{1/2}$ of the reaction

iii) The time taken for the completion of 75% reaction is thrice the $t_{1/2}$ of the reaction

iv) The pre-exponential factor in the Arrhenius equation has the dimension of time, T^{-1}

A) only (i) and (iv)
B) only (i) and (iii)
C) only (ii)
D) Only (iv)

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For More Material Join: @JEEAdvanced_2025

41. **Match List I with List II**

Column I		Column II(Type of Redox reaction)	
a)	$2Na\left[Ag(CN)_2\right]_{(aq)} + Zn_{(s)} \rightarrow Na_2\left[Zn(CN)_4\right]_{(aq)} + 2Ag_{(s)}$	I)	Combination
b)	$2Mg_{(s)} + O_{2_{(g)}} \rightarrow 2MgO_{(s)}$	II)	Disproportionation
c)	$3S_{(s)} + 6NaOH_{(aq)} \rightarrow 2Na_2S_{(aq)} + Na_2SO_{3_{(aq)}} + 3H_2O_{(l)}$	III)	Comproportionation
d)	$SO_2 + 2H_2S_{(g)} \rightarrow 2H_2O_{(l)} + 3S_{(s)}$	IV)	Displacement

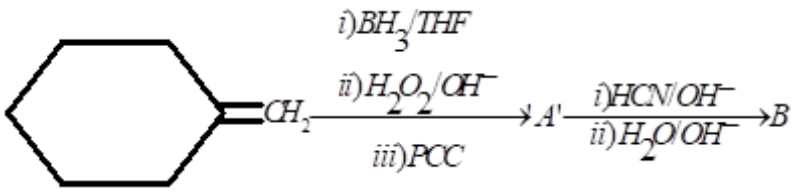
A) $a-(I);b-(II);c-(III);d-(IV)$

B) $a-(III);b-(II);c-(I);d-(IV)$

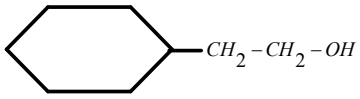
C) $a-(II);b-(III);c-(IV);d-(I)$

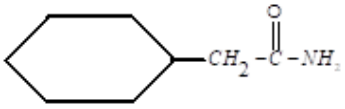
D) $a-(IV);b-(I);c-(II);d-(III)$

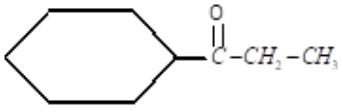
42.

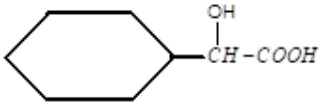


Among the following, which product comes on acidification of 'B'?

A) 

B) 

C) 

D) 

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43. Which of the following compounds is **incorrectly** named?

A) $(CH_3)_2CHCH_2CH_2CH_2CHO$ (5-Methyl-1-hexanal)

B) $(CH_3)_2CHCH_2C \equiv C - COOH$ (5-Methyl-2-hexynoic acid)

C) $CH_3CH_2CH_2CH_2CH(CH_3)COOH$ (2-Methylhexanoic acid)

D) $CH_3CH_2CH = CH - COCH_3$ (3-Hexen-5-one)

44. For a d^4 metal-ion in an octahedral field, the correct electronic configuration is

A) $t_{2g}^3 e_g^1$ when $\Delta_0 < P$ B) $t_{2g}^3 e_g^1$ when $\Delta_0 > P$

C) $t_{2g}^4 e_g^0$ when $\Delta_0 < P$ D) $t_{2g}^2 e_g^2$ when $\Delta_0 < P$

45. For the reaction $CO_{(g)} + Cl_{2_{(g)}} \rightleftharpoons COCl_{2_{(g)}}$ the k_p / k_c is equal to

A) $\frac{1}{RT}$ B) RT C) $\sqrt{RT}$ D) 1.0

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The total no of species among the following which have H-bonding

$NH_{3(g)}, H_2O_{(g)}, HF_{(g)}, H_2O_{2(l)}, CH_3COOH_{(l)}, NaHCO_3, KHCO_3$, o-fluorophenol o-nitrophenol, p-nitrophenol, Salicylaldehyde, salicylic acid, chloral hydrate, glycol, glycerol

47. The Freezing point of benzene decreases by $0.45^0 C$ when 0.2 gm of acetic acid is added to 20 gm of benzene. If Acetic acid associates to form a dimer in benzene, the percentage association of acetic acid in benzene will be----(K_f for Benzene = 5.0 $K.kg.mol^{-1}$)

48. The Entropies of X_2 , Y_2 and XY_3 are 60, 30 and 50 $JK^{-1}mol^{-1}$ respectively. For the reaction $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightleftharpoons XY_3$, $\Delta H = -30kJ$ to be at equilibrium, the temperature (in kelvin scale) should be ___

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49. The angular momentum of an electron in a certain orbit of Li^{+2} ion is 3×10^{-34} (in SI units) the potential energy of an electron in the same orbit of Li^{+2} ion is 'p' eV, thus $|p|\times10^{1}$ =____eV (Given: $E_{1_{(H)}}=-13.6eV$, Planck's constant (h) = $6.4\times10^{-34}J-\sec$, $\pi=3.2$)

50. The half-life period of certain Zero order reaction $R\rightarrow$products is 1 hr, when the initial concentration of the reactant 'R' is 2 mol/lit. How much time (in minutes) does it take for its concentration to come from 0.50 mol/lit to 0.25 mol/lit.

MATHS **MAX.MARKS: 100**

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let ABCD be a square and another square $AB_1C_1D_1$ is formed such that B_1,C_1,D_1 are mid points of AB, AC, AD respectively. Similarly, $AB_2C_2D_2$ is a square formed such that B_2,C_2,D_2 are mid points of AB_1,AC_1,AD_1 respectively. This process is continued indefinitely and the sum of areas of all squares is 12 square units, then side of square $ABCD$ is _____

A) 3 B) 4 C) 9 D) 16

52. Let $P=\{2,4,6,8\}$ and R be a relation on P defined as $R=\{(x,y):x+y\le10\}$. If m is number of elements in R and n is minimum number of elements to be added to make 'R' is equivalence then 'mn' is

A) 60 B) 48 C) 20 D) 24

53. If letters of word 'IITJEEEXAM' is arranged in all possible ways, probability that all E's are together but I's not together is $\frac{m}{n}$ then $m+n=$

A) 19 B) 47 C) 23 D) 21

54. Solution of the differential equation $\left(x\frac{dy}{dx}+y\right)=e^{xy-\ln x^2}\left(x\frac{dy}{dx}-y\right)$

A) $\frac{y}{x}-e^{-xy}=c$ B) $\frac{x}{y}+e^{-xy}=c$ C) $\frac{y}{x}+e^{-xy}=c$ D) $-\frac{x}{y}+e^{-xy}=c$

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55. In acute triangle ABC, equation of side BC is $x-y=0$. Circumcentre and orthocentre of the triangle are $(2,3)$ and $(5,8)$ respectively. Equation of circumcircle of the triangle is:

A) $x^2+y^2-4x-6y+4=0$ B) $x^2+y^2-4x-6y-4=0$

C) $x^2+y^2-4x-6y+27=0$ D) $x^2+y^2-4x-6y-27=0$

56. Value of $\underset{n\to\infty}{Lt}\dfrac{\left(\prod_{K=0}^{n+1}\dfrac{(n+1)!}{(n-K+1)!K!}\right)\left(\prod_{K=0}^{n-1}\dfrac{(n-1)!}{(n-K-1)!K!}\right)}{\left(\prod_{K=0}^{n}\dfrac{n!}{(n-K)!K!}\right)^2}$

A) $\sqrt{e}$ B) e C) e^{-2} D) $\dfrac{1}{\sqrt{e}}$

57. Sum of coefficients of all terms containing integral power of x in expansions of $\left(1-2\sqrt{x}\right)^{20}$ is ____

A) $3^{20}-1$ B) $\dfrac{3^{20}+1}{3}$ C $\dfrac{3^{20}-1}{2}$ D) $\dfrac{3^{20}+1}{2}$

58. If both the roots of the equation $x^2+ax+b+1=0$ are positive integers, then a possible value of a^2+b^2 is

A) 50 B) 37 C) 61 D) 27

59. If $|z|=1$ and $z\neq 0$, then the points $z+\dfrac{1}{z}$ lie on a

A) Straight line not through the origin B) Circle

C) Line segment lying on the x-axis D) Line segment lying on the y-axis

60. If $y=\dfrac{x+c}{1+x^2}$; c is a constant, has an extreme value at $x=x_0$, and the value of $x_0.y(x_0)=\lambda$, then $\dfrac{25}{\lambda^2}=$

A) 25 B) 100 C) 50 D) 200

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61. Let $\vec{r}$ be a variable vector and vector equations $\vec{r}\times\vec{a}=\vec{b}$ and $\vec{r}\times\vec{c}=\vec{d}$ where $\vec{a},\vec{b},\vec{c},\vec{d}$ are non-zero vectors, are consistent then, which of the following need not be true

A) $\vec{a}.\vec{b}+\vec{c}.\vec{d}=0$ B) $\vec{b}.\vec{c}+\vec{a}.\vec{d}=0$ C) $\vec{a}.\vec{b}-\vec{c}.\vec{d}=0$ D) $\vec{a}.\vec{c}+\vec{b}.\vec{d}=0$

62. The letters of the word "MANKIND" are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is

A) 1942 B) 1492 C) 1490 D) 1491

63. If $f(x)=(2x-3\pi)^{25}+\frac{4}{3}x+\cos x$ and $g(x)$ is inverse of $f(x)$ then find $\frac{d}{dx}(g(x))$ at $x=2\pi$

A) $\frac{7}{3}$ B) $\frac{3}{7}$ C) $\frac{30^{25}+4}{3}$ D) $\frac{-3}{7}$

64. The area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

A) ab B) 2ab C) $\frac{a}{b}$ D) $\sqrt{ab}$

65. If L_1 is line joining points $\left(\frac{5}{2},0\right)$ and $(0,-5)$, and L_2 is line joining points $\left(\frac{1}{2},\frac{1}{2}\right)$ and $(0,1)$ and $L_3: 3x-y-k=0$ are concurrent lines then k is

A) 5 B) 3 C) 9 D) 7

66. The range of $\cos\theta\left(\sin\theta+\sqrt{\sin^2\theta+\sin^2\alpha}\right)$ is

A) $\left[-\sqrt{1+\sin^2\alpha},\sqrt{1+\sin^2\alpha}\right]$ B) $\left[-\sqrt{1+\cos^2\alpha},\sqrt{1+\cos^2\alpha}\right]$

C) $\left[-\sqrt{1+\cos^2\alpha},\sqrt{1+\sin^2\alpha}\right]$ D) $\left[-\sqrt{1+\sin^2\alpha},\sqrt{1+\cos^2\alpha}\right]$

67. Let $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ and $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$. If $\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right).\hat{k}=\frac{23}{2}$, then $\left|\vec{b}\times 2\hat{j}\right|$ is equal to

A) 4 B) 5 C) $\sqrt{21}$ D) $\sqrt{17}$

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68. Area (in sq. units) of the region outside $\frac{|x|}{2}+\frac{|y|}{3}=1$ and inside the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ is:

A) $6(\pi-2)$ B) $3(\pi-2)$ C) $3(4-\pi)$ D) $6(4-\pi)$

69. $\int \frac{x^2-1}{(x^2+1)\sqrt{x^4+1}}dx=$

A) $\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)+c$ B) $\frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)+c$

C) $\frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}}\right)+c$ D) $\sec^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+c$

70. If A is a matrix of order $n\times n$, $\left|adj\left(adj\left(adj\left(A\right)\right)\right)\right|=|A|^{(n-1)^k}$ then $k=$

A) 4 B) 6 C) 3 D) 7

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Number of points where the function $f(x)=[x]+\left[x+\frac{1}{4}\right]+\left[x+\frac{1}{2}\right]+\left[x+\frac{3}{4}\right]$ (where $[.]$ denotes greatest integer function) is discontinuous in (0,4) is_____

72. If $\lim\limits_{x\to 0}\int_0^x \frac{t^2dt}{(x-\sin x)\sqrt{a+t}}=1$, then value of 'a' is

73. The probability distribution of random variable X is given by:

X	1	2	3	4	5
$P(X)$	K	$2K$	$2K$	$3K$	K

Let $R=P\left(1<X<4\middle|X<3\right)$. If $5R=\lambda K$, then λ equal to

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74. Let $\frac{x}{\lambda}=\frac{y}{2}=\frac{3(z-1)}{7}$ and $\frac{x+1}{-1}=\frac{y}{3}=\frac{z}{1}$ are two skew lines; $\lambda \in N$. If shortest distance between them is $\frac{K}{\sqrt{970}}$ for minimum value of λ then K is

75. If a and b are real and $ax+(\sin b)y+(\cos b)z=0; x+(\cos b)y+(\sin b)z=0;$ $-x+(\sin b)y-(\cos b)z=0$ then number of integral values of 'a' for which the system of linear equations has a non-trivial solution is

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